

Best practices for reducing pain associated with intrauterine device placement



Lisa L. Bayer, MD, MPH, FACOG; Samir Ahuja, MD, FACOG; Rebecca H. Allen, MD, MPH, FACOG; Melanie A. Gold, DO, DMQ, DABMA, FAAP; Jeffrey P. Levine, MD, MPH; Lynn L. Ngo, MD, MPH, FACOG; Sheila Mody, MD, MPH, FACOG

Intrauterine devices are highly effective, long-acting, reversible forms of contraception. Despite their benefits, limited uptake persists, particularly among underserved populations, adolescents, and young adults. While most intrauterine device placement procedures are uncomplicated, pain is commonly reported, with more severe pain reported in certain groups. No current standard of care has been established specifically to manage pain with intrauterine device placement, resulting in wide variation in clinical practice. This article aims to provide practical, evidence-based, and expert-informed guidelines for managing pain during intrauterine device placement. The authors (a group of board-certified obstetrician-gynecologists, pediatricians, and a family physician) conducted a virtual expert meeting to develop consensus-based recommendations for pain management. The meeting covered environmental considerations, nonpharmacological, and pharmacological options. Key strategies include using a person-centered care model that focuses on patient values, needs, and preferences to promote shared decision-making around pain relief. Clinicians should assist patients in setting realistic expectations, including a discussion of the placement procedure, anticipated pain, and all options for pain relief, to allow for a comprehensive informed consent process. Nonpharmacological interventions, such as the use of therapeutic language and comforting environmental, complementary and integrative, and mind-body elements, are recommended to reduce patient anxiety and discomfort. In terms of pharmacological options, clinicians may offer pre-procedural—specific nonsteroidal antiinflammatory drugs (eg, naproxen and ketorolac) and anxiolytics to patients with significant anxiety, as well as intraprocedural application of topical anesthesia and paracervical or intracervical blocks. Finally, additional techniques, such as using appropriately sized instruments and ensuring gradual and gentle procedural steps, can also help enhance patient comfort during insertion. These expert guidelines emphasize the importance of personalized, trauma-informed, and evidence-based care, prioritizing patient autonomy and preferences to facilitate a safe and acceptable insertion experience. They are critical to reducing barriers to intrauterine device uptake and improving overall patient outcomes.

Key words: analgesia, anxiety/prevention & control, contraception, integrative medicine (preferred MeSH term), intrauterine device (IUD), pain/prevention & control, paracervical block

Introduction

Intrauterine devices (IUDs) are highly effective, long-acting reversible contraceptives providing contraceptive

coverage for up to 3 years (levonorgestrel 13.5 mg IUD [LNG-IUD]), 5 years (19.5 mg LNG-IUD), 8 years (52 mg LNG-IUD), or 10 years (copper

380 mm² IUD). However, there is underutilization, particularly among underserved populations, adolescents, and young adults, largely due to structural

From the Division of Complex Family Planning, Department of Obstetrics and Gynecology, Oregon Health & Science University, Portland, OR (Bayer); Department of Obstetrics & Gynecology, University Hospitals, Cleveland, OH (Ahuja); Department of Obstetrics and Gynecology, Warren Alpert Medical School of Brown University, Providence, RI (Allen); Special Lecturer, Department of Pediatrics and Department of Population & Family Health, Columbia University Irving Medical Center, New York, NY (Gold); Professor and Director of Reproductive & Gender Health Programs, Department of Family Medicine and Community Health, Rutgers Robert Wood Johnson Medical School, New Brunswick, NJ (Levine); Department of Obstetrics and Gynecology, Southern California Permanente Medical Group, San Diego, CA (Ngo); Director, Division & Fellowship Complex Family Planning, Department of Obstetrics, Gynecology & Reproductive Services, UC San Diego, CA (Mody).

Received Sept. 2, 2024; revised Jan. 21, 2025; accepted Jan. 28, 2025.

All authors participated in a virtual advisory board meeting supported by Bayer. Following the advisory board, the authors developed the concepts and manuscript. The authors were not paid for their participation in the writing process. Melanie Gold: Consultant presenting on Patient Centered Contraceptive Counseling (not product-specific; current): Bayer. Research funding on Telehealth-Supported Long-Acting Reversible Contraception in School Based Health Centers (ended 2023): Merck/Organon. Consultant on expert panel on Scientific Workshop on Shared Decision Making (ended 2023): Glaxo Smith Kline (GSK). Research Consultant for Policy and Research Group: Current. Research Consultant for 5 Point App: Current. Consultant for The Tapping Solution, LLC and The Tapping Solution Foundation: (Ended August 31, 2024). Rebecca Allen attended a Bayer Advisory Board for a new IUD inserter (2024). The remaining authors report no conflict of interest.

A virtual advisory board meeting was held on March 7, 2024, supported by Bayer, who also funded medical writing and editorial support.

Corresponding author: Lisa Bayer, MD, MPH. bayerl@ohsu.edu

0002-9378 • © 2025 The Authors. Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>). • <https://doi.org/10.1016/j.ajog.2025.01.039>

factors.^{1–3} While most IUD placements are uncomplicated, pain during placement is commonly reported.⁴ Studies have shown that a higher proportion of nulliparous patients experience pain with IUD placement than those who are parous,^{5–8} with one study reporting severe pain in 67.2% of nulligravida, 47% with a history of vaginal delivery, and 42% with prior cesarean section.⁷

Pain and fear of pain related to IUD placement represent notable barriers to IUD uptake.^{9,10} Social media platforms have recently brought attention to concerns about negative experiences with IUD placement and the potential for insufficient informed consent, which can lead to patient discomfort and contribute to broader issues of medical distrust.^{11–14} Because anxiety and anticipatory pain correlate with higher reported pain intensity,^{15,16} these negative public perspectives may contribute to undesirable perceptions.^{11–14} Compounding this issue, clinicians tend to underestimate the pain associated with IUD placement and, therefore, risk undertreating pain.^{4,17} Currently, there is no established standard of care addressing pain with IUD placement,¹⁸ leading to wide variation in clinical practice.¹⁹ The United States Centers for Disease Control and Prevention recently released updated guidance on the provision of medication for IUD placement in the 2024 U.S. Selected Practice Recommendations for Contraceptive Use, emphasizing the need to address pain and providing evidence-based recommendations.²⁰

This article aims to present practical, evidence-based, expert-informed guidance for clinicians to optimally manage pain surrounding IUD placement. We emphasize the importance of trauma-informed care, person-centered counseling, and an optimal clinical environment, while offering recommendations for nonpharmacological and pharmacological interventions. Some options suggested below may not be available in all clinical settings; however, clinicians lacking access to specific interventions should not be deterred from offering IUDs where clinically indicated.

Methodology

A virtual advisory board meeting was held on March 7, 2024, supported by Bayer (Whippany, NJ, United States), manufacturer of the LNG-IUD (Mirena, Kyleena, and Skyla/Jaydess) and a copper-releasing IUD (Nova T) (Jaydess and Nova T are not available in the U.S). Participants included 6 practicing board-certified obstetrician/gynecologists, 2 adolescent medicine subspecialist pediatricians, and a family medicine physician, all with extensive experience in IUD placement and an interest in pain management. Four are complex family planning subspecialists; 4 are clinical trainers; one is a member of a guidance panel related to IUD placement; one has contributed to IUD guidelines; and 4 have published research on pain during IUD placement.

The participant briefing included the following assumptions: there were no contraindications to IUD placement; informed consent for IUD placement was given; and choice of IUD was agreed upon. The experts reviewed the published evidence, shared their clinical experiences, and developed recommendations for managing discomfort during IUD placement. Recommendations were collated and developed over multiple writing iterations. Evidence was based on relevant randomized controlled trials (RCTs), systematic reviews, and meta-analyses; where these were unavailable, comparable studies were used, predominantly for related gynecological procedures. Where evidence was lacking, low quality, or contradictory, expert recommendations were made based on clinical experience.

Recommendations

Overarching principles

When performing IUD placement, key principles include respect, patient autonomy, informed consent, shared decision-making, and trauma-informed, personalized care (Figure 1). Patients should feel empowered, informed, and in control; upon request, the procedure should be paused, stopped, or rescheduled, at any stage, and an

alternative contraceptive provided if desired.

Staff training and service philosophy

An optimally trained team can improve the patient's journey by reducing misinformation, error, anxiety, and pain. Training administrative and clinical support staff (eg, medical assistants, nurses) is essential to provide patients with accurate information, instill trust, and inspire confidence.²¹ Common misinformation that can result in unnecessary delays in IUD placement include recommendations that the patient must be menstruating or that a separate visit is required for sexually transmitted infection and cervical cancer screening before IUD placement. Of note, a review by Whiteman et al found no association between the timing of IUD placement with menses and reduced insertion pain.²² Table 1 outlines important practical information that administrative staff can share with patients. Sufficient time should also be scheduled for the IUD placement, including patient counseling and consent process, procedure duration, and aftercare. For some patients, this may require 2 visits.

In-procedure clinical support staff (eg, medical assistants, nurses) should be trained in recognizing and managing signs of impending vasovagal events (pale skin, sweating, dilated pupils; complaints of dizziness, nausea) and discomfort and communicating these to the clinician. A trained chaperone who directly observes the patient's experience of comfort, dignity, and professionalism is recommended during sensitive exams.²³

Trauma-informed care

Trauma-informed care (TIC) suggests that exposure to abuse, neglect, discrimination, violence, and other adverse experiences increases a person's lifelong potential for serious health problems and health-risk behaviors.²⁴ TIC further acknowledges the need to understand a patient's life experiences to deliver effective care and improve engagement, treatment adherence, and

FIGURE 1

Recommendations for training, service philosophy, counseling, and language

Recommendations to optimize IUD placement comfort

Overarching principles

- **Respect, empowerment, informed consent, and choice:** The individual is informed and in charge
- **Shared decision-making:** Decisions about clinical care are made in equal partnership
- **Permission:** The clinician checks in with the individual periodically; the individual can stop the procedure at any time
- **Personalized care:** An individualized approach based on prior experiences, needs, and preferences
- **Eliciting and valuing each individual's experiences:** Care is based on what works and does not work for the individual
- **Therapeutic alliance and team approach:** Requires a trained and competent team from first to last contact
- **Trauma-informed care:** Consider past trauma; avoid triggering language; manage own body language and behaviors

Staff training and service philosophy

Administrative staff (scheduling, reception) are trained to provide accurate information (also see Table 1):

- In most cases, IUD counseling, consent, and placement can be performed in **one visit**
- Schedule insertion anytime during **menstrual cycle**
- Schedule **sufficient time** for counseling and the procedure
- Advise patient to **eat and drink** beforehand
- Encourage appropriate **clothing** and **premedications** (see Figure 2)
- Provide pre-visit patient **resources**
- Utilize empowering, **therapeutic language**

Clinical support staff:

- **Trained** in the use of **therapeutic language** and their role as **supporter** at the patient's head
- Preparation of **equipment**
- Monitoring and ordering **supplies**
- Facilitating a **smooth process**
- Providing **support** during the procedure
- Being aware of impending **vasovagal episode** and how to respond

Person-centered counseling

Approach:

- **Control:** The patient is in control at all times; they can pause or stop the procedure at any time, for any reason
- **Body language and positioning:** Clinician lower than patient; elevate exam table so patient can see clinician at all times; open body language; confident, approachable
- **Expectations:** Set personalized realistic expectations
- **ASA (Affirm, Share, Ask) approach:** Affirm the individual's query, share information, ask a follow-up question using active listening techniques
- **Teach re-focusing techniques:** Notice support of exam table, crinkling of clean white paper; slow, gentle breathing; breathe in comfort and exhale discomfort and tension
- **Clothing advice:** Individual is dressed during counseling and consent process; wears comfortable clothing with the ability to remove bottoms and keep top intact

Information:

- **Describe procedure** using layperson terms, including each step from start to finish and how long it lasts
- There are three steps during the procedure when discomfort may be experienced: placement of cervix holder (tenaculum), measuring the uterus (sounding), and gently placing the IUD
- Compare discomfort to something patients are familiar with (e.g., menstrual cramps); may use a scale of 0–10
- **Explain** that patients might feel nothing or other sensations (pressure, pulling, nausea, cramping)
- **Show:** Instruments and devices (as appropriate)
- **Analgesic options** including NSAIDs, local anesthetics, paracervical block for office-based placement, moderate sedation, or deep sedation in OR (if available; see Figure 2)
- **Anxiolytic options:** benzodiazepines require informed consent for IUD placement before drug administration and someone to take them home
- **Aftercare and management** post-procedure
- **Written patient instructions** at the time of counseling*

Therapeutic language and verbal analgesia

- **Expectations:** "This procedure shouldn't cause you any harm. Patients may have discomfort, but everyone's experience is different, and we have ways to help make this more comfortable. We want this to go to well for you. If you need a different plan to help with any discomfort, let me know right away. There are three steps where you might have sensation/possible discomfort; we will pause after each step until you let us know that you are ready to continue."
- **Control:** Ask permission to start. "You are in control. We can stop at any time. We can reschedule if now is not good for you." **
- **Word choice:** "Gentle placement" (not insertion); "cervix holder" (not tenaculum); "measuring your uterus" (not "uterine sound/dilator"); "exam table" (not bed); "check/examine" (not inspect/look at/touch); "you may feel a sensation" (not stick/burn/prick).
Example sentences: "It's a natural reaction to lift up; see if you can let your hips be heavy on the table"; "I am going to place the speculum".
- **Trauma-informed:** Awareness; use patient-centered language and avoid triggering language, e.g., "let your knees fall out towards the walls"

*If not same-day insertion. **Discuss backup contraception.

IUD, intrauterine device; NSAID, nonsteroidal anti-inflammatory drug; OR, operating room.

health outcomes.²⁴ Universal screening for trauma should be performed before any pelvic exam or procedure,²⁵ and the

principles of TIC should be applied to all encounters to develop a safe space for patients. Both administrative and

clinical staff should be trained in trauma-informed practices and optimal use of therapeutic language.

TABLE 1**Practical information for administrative and support staff to share with individuals inquiring about or scheduled for IUD placement**

The procedure usually takes 5–10 min, and we book additional time for comfort and to ensure all your questions are answered (total time depends on local protocols).

In most cases, only one visit is needed: all necessary testing can be done in one visit (such as screening for pregnancy, sexually transmitted infections, and cervical cancer).

You can have an IUD placed whether you are menstruating or not, as long as it is reasonably sure you are not pregnant.^{a,b} There is no best time in the menstrual cycle for IUD placement.²²

The copper 380 mm² IUD and the LNG 52 mg IUD can be used as emergency contraception (EC) up to 5 days (copper IUD^b) or 7 days (LNG 52 mg IUD^c) after unprotected intercourse, although LNG 52 mg IUD for EC remains subject to expert debate.^{122,123}

Most patients tolerate IUD placement well. You will be able to talk to the clinician about your options for managing discomfort. Some of these options will require additional time and preparation or a separate appointment.

If the patient desires same day placement and has no contraindications, provide prescription for naproxen 500–550 mg PO or ketorolac 20 mg PO 1 h before the procedure (or 2 over-the-counter naproxen sodium=440 mg). Other NSAIDs can be used if naproxen is unavailable. Always take NSAIDs with food.

You should eat a light meal (for blood sugar) and drink water (for hydration) beforehand.

You should dress comfortably and normally. It can be helpful to wear a separate top and bottom so you can just remove the bottom for the procedure.

Some patients may experience discomfort or dizziness after IUD placement, which usually settle after resting for a short time.

You can decide whether you want to return to work/school/other activities after the procedure or take some time off.

Some patients find it helpful to have a support person accompany them to the visit or a ride home after the placement.

CDC, Centers for Disease Control and Prevention; IUD, intrauterine device; LNG, levonorgestrel; NSAID, non-steroidal anti-inflammatory drug; PO, orally; SPR, Selected Practice Recommendations.

^a Determining pregnancy status is outside the scope of this paper; interested practitioners may refer to the CDC SPR guidance available online: (https://www.cdc.gov/mmwr/volumes/73/rr/rr7303a1.htm?s_cid=rr7303a1_w#howtobereasonablycertainthatapatientisnotpregnant); ^b If the day of ovulation can be estimated, the Cu-IUD may be placed >5 days after intercourse as long as placement does not occur >5 days after ovulation; no additional contraceptive protection is indicated after Cu-IUD placement.¹¹⁷; ^c If the LNG-IUD is placed within the first 7 days since menstruation started, no additional contraceptive protection is indicated. If the LNG-IUD is placed >7 days after menstrual bleeding starts, the patient should abstain from intercourse or use barrier methods for the next 7 days.¹¹⁷.

Person-centered counseling and shared decision-making

Person-centered counseling, shared decision-making, and informed consent empower patients to make informed choices and improve patient satisfaction.^{20,26,27} The expert panel recommends clinicians move away from directed counseling and medical paternalism and agree on a patient-centered plan for IUD pain management based on individual preferences (Figure 1).²⁰

Part of a person-centered approach includes establishing mutual trust and respect when addressing pain expectations. While negative pain expectations increase patients' pain perceptions,^{15,28} clinicians should take care not to minimize pain, which can lead to feelings of mistrust and betrayal.²⁹ A candid discussion about anticipated pain should consider risk factors for a more painful procedure (Table 2), as well as the concerns and emotional needs of the patient.³⁰ Analgesia options should be discussed and offered to all patients,

irrespective of risk factors for increased pain. The benefits and risks of available options and alternatives should be addressed, and a holistic pain management strategy should honor individual needs, preferences, and values.²⁰

Identify risk factors for a more painful procedure

The perception of pain is multifaceted and extends beyond the activation of nociceptors. Pain may be experienced at any anatomical level of the IUD placement: (1) central nervous system: anticipatory anxiety, fear of pain, history of trauma, psychiatric conditions; (2) vulvovaginal: pelvic exam with bimanual exam and speculum placement; (3) cervical: applying the tenaculum or another instrument to stabilize the cervix and traversing the cervix; (4) uterine: cramping with uterine sound and IUD contacting the fundus.³¹ When counseling, patients should be assessed for factors that increase the risk of a painful procedure (Table 2).¹¹

The type of IUD placed may also contribute to variations in pain. Patients receiving devices with a smaller diameter of inserter (13.5 mg or 19.5 mg LNG-IUD) reported less pain than those receiving the 52 mg LNG-IUD.^{6,32} Some studies have shown increased pain from the 52 mg LNG-IUD compared to the copper 380 mm² IUD, while other studies have not seen a difference.^{7,16,33,34} For most patients, other profile differences between the IUDs (eg, duration of use, bleeding patterns, hormonal/nonhormonal IUD) are the most important when considering device type,^{7,16} but some may also factor in the differences in pain during placement.

Informed consent

Medical exams and procedures must be explained and performed with the patient's informed consent.²⁷ The patient should be provided with sufficient, accurate, and understandable information. They should have the capacity to process this information and the freedom to ask

questions and make voluntary decisions.²⁷ Clinicians may offer to show the speculum and IUD so patients can visualize the devices involved.^{27,35}

Patients manage pain better when they are well-counseled on what to expect during the procedure.³⁶ Patients should be prepared for 3 sensation points after the pelvic exam and speculum placement: (1) the application of the instrument to stabilize the cervix, (2) the sounding of the uterus, and (3) the IUD placement through the cervix into the uterus. The clinician should explain that they will wait for each sensation to diminish and proceed only when the patient is ready. Clinicians may compare the level of discomfort to experiences that patients are familiar with (eg, menstrual cramps) using a scale of 0 (no discomfort) to 10 (worst pain ever). Notably, while most patients reported mild to moderate pain even without the use of analgesia,^{4,8,37} pain can be greater among those with additional risk factors (Table 2). For example, nulliparous patients in trials with placebo arms reported a higher median pain score of 6.6 to 8.1/10 on the visual analog scale compared to multiparous patients (3.7/10).^{38,39}

Pain and anxiety management options (see Figure 2 and below) should be discussed with patients prior to the procedure, with the latter particularly important for anxious patients or those with a history of trauma. In cases of severe anxiety, history of sexual trauma, complicated anatomy, intellectual disability, patient preference, or other indications, referral to an appropriate facility for sedation or placement under anesthesia may be appropriate.⁴⁰ Patients should also be informed about what to expect after the procedure, recommended analgesics, self-care, expected levels of pain, cramping, bleeding and its management (eg, use of tampons or menstrual cups), and the duration for which they may require backup contraception (Figure 2). If the placement becomes more complex, continue to check in with the patient about proceeding or rescheduling with a different pain management plan.

TABLE 2

Factors associated with the likelihood of increased pain during IUD placement^a

Social, demographic, and psychological factors	Physical and medical factors
Age (adolescence)	Nulliparity
History of trauma	Multiple cesarean deliveries
Anxiety or mood disorder	Not currently breastfeeding
Baseline anxiety (fear)	Dysmenorrhea
Anticipation or expectation of pain	Anatomical: extreme retroverted uterus, anteфлекed/retroflexed uterus, narrow cervical os, tortuous cervical canal, uterine fibroids
Previous painful IUD placement	Prior cone biopsy of cervix
Previous negative reaction to pelvic exam	Prior failed IUD placement
Race ^b	Size of IUD inserter ^c
Lack of mental preparation	Difficulty or pain with uterine sound
Higher level of education (≥ 7 y)	Time between last delivery and IUD placement (>13 mo)
Higher emotional reactivity	Menstruation (nulligravidas)

IUD, intrauterine device; LNG, levonorgestrel.

^a Currently, it is not possible to predict with certainty whether a patient will experience severe pain or discomfort during the procedure; ^b Race as a risk factor is likely due to complex social and institutional realities and inadequately treated pain; ^c Increased pain has been reported with 52 mg LNG-IUD compared to 13.5 mg, 19.5 mg, or copper 380 mm² IUD [Table adapted from Gemzell-Danielsson et al³⁸ with additional sources^{7,16,31,33,34,124–127}].

Nonpharmacological approaches

Given the direct relationship between anxiety and pain perception, nonpharmacologic approaches that alleviate anxiety may help reduce pain associated with IUD placement.⁴¹ Clinicians are encouraged to use therapeutic language, support patient choice, and create an optimal environment using nonpharmacological/complementary and integrative medicine (CIM) interventions. The lack of an optimal environment should not limit access to IUD provision. Depending on the patient's circumstances, a quick and safe placement of a reliable, long-acting reversible contraception may outweigh any environmental limitations.

Language

"Verbal analgesia" ("vocal local" or "verbocaine") involves calming, comforting, and distracting patients throughout the procedure using a soft, low-pitched voice and speaking slowly.^{42,43} This may include casual talk to distract the patient or promote positive expectancy for comfort, confidence,

and an easy and quick procedure. In one study, verbal analgesia demonstrated comparable analgesic effects to oral tramadol during placement of the 13.5 mg LNG-IUD among nulliparous patients.⁴⁴

Studies evaluating related procedures have demonstrated the importance of word choice. During speculum exams, patients hearing phrases with unpleasant connotations ("you are going to feel a lot of pressure") reported significantly higher pain scores than those hearing objective phrases ("I am going to introduce the speculum").⁴⁵

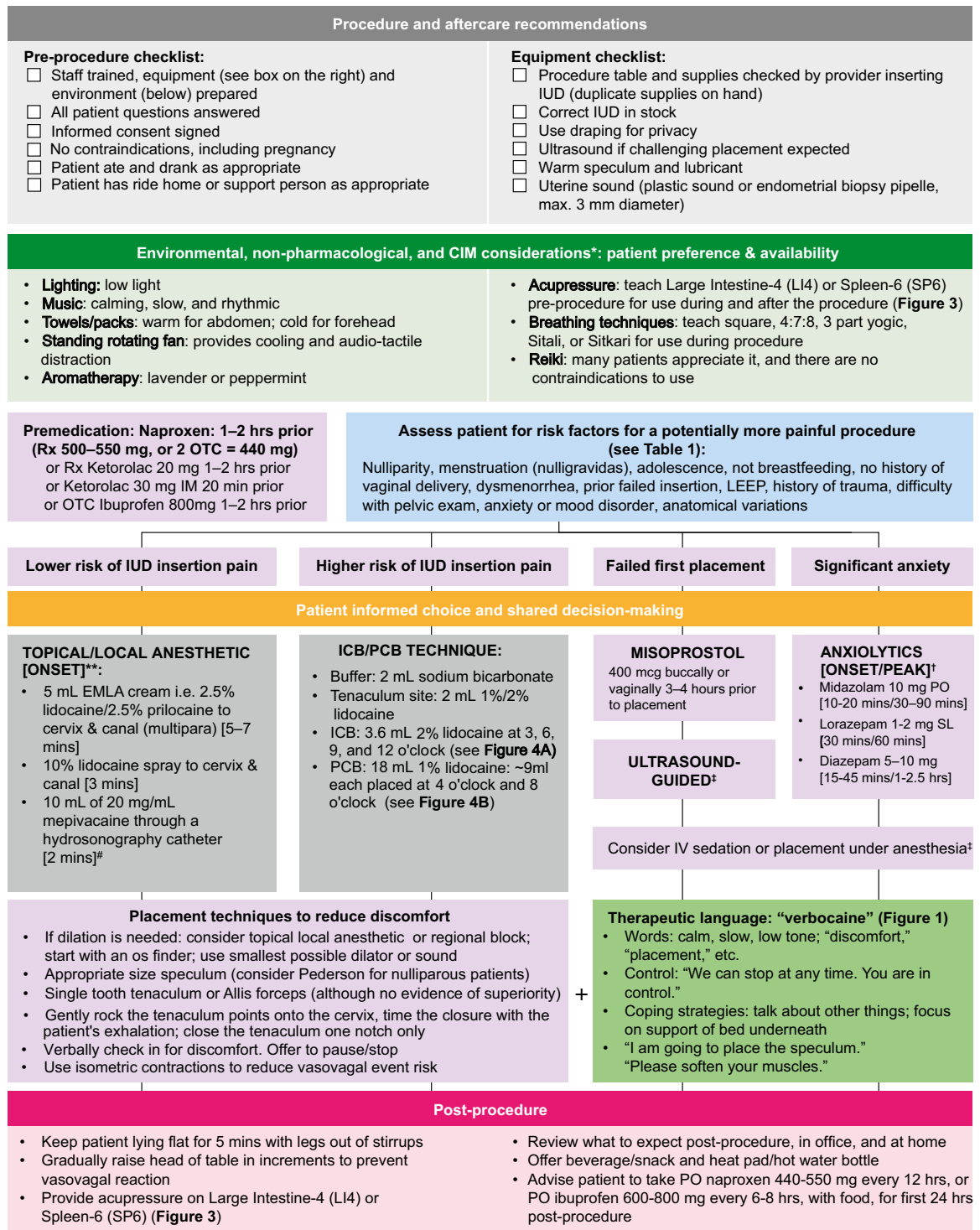
In addition, the expert group recommends avoiding medical jargon or negatively loaded language when talking to patients. Examples of suggested language are given in Figure 2.

An optimal environment

Simple strategies (Figure 1), such as low room lighting, soothing music, and a cooling fan, can help optimize the clinical environment.^{46–48} Warm towels or heated pads placed on the lower

FIGURE 2

Practice recommendations for minimizing pain with IUD placement

 Overarching principles: respect, empowerment, and choice
 Is discomfort tolerable? Ask permission at each step


*Can be offered to patients but there is a limited evidence base. **Wait times based on onset time are given in square brackets. Availability of topical anesthetics may vary by country. †Onset, onset of action; peak, peak blood levels (given in square brackets). Benzodiazepines reduce anxiety, not pain. If a benzodiazepine is prescribed, the patient cannot drive afterward and will require a person to take them home. Note that simultaneous use of benzodiazepines and opioids may result in respiratory depression, sedation, coma, and death.⁸² ‡Only if available; not covered in this guidance. #Note that hydrosonography catheters may not be widely available in clinics where IUD placements occur. Alternatives may include an 18-gauge angiocatheter or 3 mm Novak curette.^{119–121}

CIM, complementary and integrative medicine; EMLA, eutectic mixture of local anesthetics; IM, intramuscular; IUD, intrauterine device; IV, intravenous; LA, local anesthetic; LEEP, loop electrosurgical excision procedure; LNG-IUD, levonorgestrel intrauterine device; OTC, over-the-counter; PCB, paracervical block; PO, oral.

abdomen or back may reduce cramping pain (based on dysmenorrhea studies),⁴⁹ whereas cold, wet towels on the forehead can provide comfort. Music, particularly “slow-rhythm music,” has been found to reduce pain and anxiety during hysteroscopy and colposcopy procedures and can be tailored to the patient’s desires.^{50–52} In one author’s practice, patients are offered free affirmation audio recordings, both before and during the procedure, on topics such as “I want my body to feel comfortable” and “I want to feel calm during procedures” (free audio resources are available from: <https://imaginaction.stanford.edu>). Several recent studies have also looked at the use of virtual reality glasses during IUD placement, but with mixed results.^{53–55}

CIM considerations

Aromatherapy with lavender^{56,57} may increase satisfaction and reduce pain or anxiety associated with IUD placement as well as other nongynecologic procedures.^{58,59} Peppermint essential oil may reduce postoperative nausea⁶⁰ and has been shown to decrease symptoms of dysmenorrhea.⁶¹ Acupuncture and acupressure have limited supportive evidence in IUD placement; however, several authors use these techniques to reduce postplacement cramping pain. A single RCT supports the bilateral stimulation of the acupressure point Large Intestine-4 (LI4, also called Hegu or Joining Valley; [Figure 3, A](#)) during IUD placement.⁶² Spleen-6 (SP6, also called three-yin intersection or Sanyinjiao; [Figure 3, B](#)) is effective for the relief of primary dysmenorrhea.^{63–65} Patients can also perform LI4/SP6 self-stimulation at home to reduce cramping. Additionally, Reiki is well-liked by patients even though the evidence is limited to hospitalized patients undergoing other gynecological procedures.⁶⁶ Mind-body techniques, such as guided imagery or hypnotic techniques, may help with overall relaxation (eg, “focus on the clean crinkling of the paper beneath you and the comfortable support of the exam table”).⁶⁴

Patients may benefit from directed coaching on yoga-based breathing during IUD placement. These include square (“box”) breathing, 4-7-8 breathing, and pranayama (yogic) breathing

techniques such as 3-part yogic, Sitali, or Sitkari breathing (for details, see <https://www.mondaycampaigns.org/destress-monday/deep-breathing-practices>).^{64,67,68} To reduce the risk of vasovagal reaction, patients can be taught to use isometric contractions of the muscles of the arms, hands, feet, and legs to reduce venous pooling.⁶⁹ Additionally, the patient can hold a stress ball or grip 2 of the supporting person’s fingers.

These CIM approaches are considered reasonable nonpharmacologic options given their low risk of adverse effects.

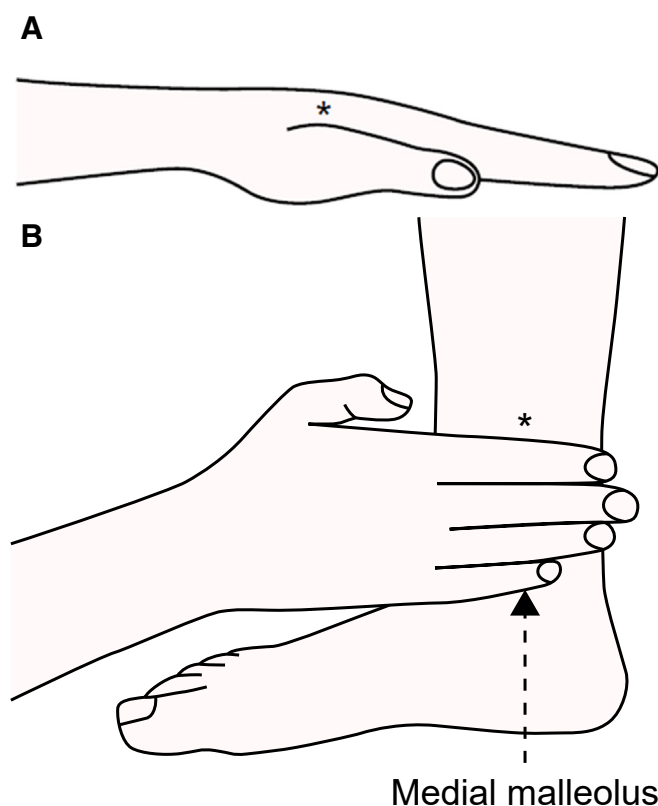
Preparation of equipment

Several practical steps are essential for optimal IUD placement (see Checklist

and Equipment boxes, [Figure 2](#)). The equipment must be checked and ready; speculums (of appropriate sizes) and lubricating gel should be available and, if possible, warmed to body temperature.⁷⁰ An atraumatic vulsellum or Littlewood forceps, compared to a single tooth tenaculum, showed no difference in pain during IUD placement.^{71,72} Options for uterine sounding include a plastic sound or endometrial biopsy pipelle and a metal sound (maximum 3 mm diameter). Consider an os finder, single-use plastic sound dilator, or a set of progressive metal dilators (eg, 13/15 Pratt uterine dilator or mini cervical dilator set) if cervical dilation is needed. Typically, dilation to 3 mm is needed for the 13.5 and 19.5 mg LNG-IUD and 5 mm

FIGURE 3

A, Diagram of acupressure point “Large Intestine-4” (LI4). B, Diagram of acupressure point “Spleen-6” (SP6)



A, Diagram of acupressure point “large intestine-4” (LI4). *LI4 acupressure point is found on the dorsum of the hand, approximately at the midpoint of the second metacarpal bone, in the belly of the first interosseus dorsalis muscle. **B, Diagram of acupressure point “spleen-6” (SP6).** *SP6 acupressure point is found approximately 4 fingers superior to the most superior point of the medial malleolus.

for the 52 mg LNG-IUD and copper IUD. However, more dilation may be required for navigating the angle between the cervix and the uterus with the plastic IUD applicator in cases of severe ante- or retroflexion of the uterus. While routine ultrasound guidance for IUD placement is not necessary or available in many clinics, it has demonstrated reduced pain compared to nonguided placement⁷³ and may be reserved for situations where cervical dilation is required.

Analgesics

Studies of nonsteroidal antiinflammatory drugs (NSAIDs) taken before IUD placement have yielded variable results in reducing discomfort during and after the procedure.^{74–76} Oral naproxen 500 mg demonstrated a reduction in pain scores during IUD placement compared to placebo in one trial of multiparous subjects⁷⁷ but not in nulliparous subjects.³⁹ The latter trial did, however, observe a clinically significant reduction in pain scores at 5 and 15 minutes after IUD placement. Oral ketorolac 20 mg, taken 40 to 60 minutes before IUD placement, reduced pain during and 10 minutes after placement.⁷⁸ Intramuscular ketorolac 30 mg given 30 minutes prior did not reduce pain with IUD placement but did reduce pain at 5 and 15 minutes after placement. Oral ibuprofen taken as premedication has not been shown to reduce pain associated with IUD placement but may be beneficial in managing pain after placement, especially since peak levels of ibuprofen typically occur 1 to 2 hours after oral administration.^{11,76,79–81}

Further, oral tramadol 50 mg 1 hour prior to IUD insertion significantly decreased pain compared to placebo and was more effective than naproxen in one study.⁷⁷ However, tramadol does require pre-planning before administration and, as an opioid, should not be given with a benzodiazepine; the patient may also need a ride to and from the appointment.

The expert group recommends prescription-strength oral naproxen 550 mg taken 1 to 2 hours beforehand or ketorolac (20 mg oral [taken 1–2 hours before] or 30 mg intramuscular [given

20–30 minutes before]) as premedication to reduce placement- and post-placement discomfort (Figure 2). Given the time restraints of NSAID administration during the IUD placement visit, self-administration of maximum dose oral over-the-counter NSAIDs (eg, naproxen 440 mg or, if naproxen is unavailable, ibuprofen 800 mg) 1 hour prior to the scheduled visit may be a practical alternative. Finally, it is important to ensure the patient has eaten prior to administration of NSAIDs to prevent gastrointestinal irritation.

Cervical priming

Misoprostol has been most commonly studied as a cervical primer prior to IUD placement. Most studies and reviews do not support the routine use of misoprostol to decrease pain with IUD placement.^{20,74,82} One study found that 200 mcg vaginal misoprostol given at 10 and 4 hours prior improved successful IUD placement for patients with a previously failed attempt.⁸³ Another study of 400 mcg vaginal misoprostol given 4 hours prior to IUD placement in nulliparous patients found decreased patient-reported procedure pain and clinician-reported placement difficulty but increased postprocedure cramping.⁸⁴ Considering its side effects and limited evidence, we recommend cautious use of misoprostol only in cases of failed placement or selected patients with risk factors for a failed insertion, for example, known cervical stenosis (see [Failed first attempt at IUD placement](#)).

Anxiolytics: benzodiazepines and nitrous oxide

There have been no studies investigating oral anxiolytics for reducing IUD placement pain. While benzodiazepines do not directly reduce pain, they can alleviate patient anxiety. Oral midazolam 10 mg taken 30 to 60 minutes beforehand is preferred due to its fast onset, short half-life (1.5–2.5 hours), and favorable safety profile (evidence based on a surgical abortion study).⁸⁵ Alternatives include 1 to 2 mg sublingual lorazepam (taken 20–30 minutes beforehand) or 5 to 10 mg oral diazepam

(taken 1–1.5 hours beforehand), both of which have longer half-lives (12 and 46 hours, respectively).⁸⁶ It is important to ensure enough time for the medication to take effect, arrange for someone to take the patient home afterward, and avoid combining it with opiates (Figure 2). Informed consent should be obtained prior to the patient taking a benzodiazepine.

Two studies have evaluated the use of nitrous oxide (N₂O) for IUD placement with mixed results. One study of nulliparous adolescents using N₂O titrated to a maximum of 70/30 (N₂O/O₂) showed an improvement in pain and satisfaction, while another study of nulliparous patients using a fixed dose of 50/50 (N₂O/O₂) showed no difference in pain but improved satisfaction.^{87,88} Higher concentrations of N₂O with titration tailored to an individual's needs may be more effective. While the use of N₂O is not widely available in all clinics, its rapid onset, safety profile, and quick recovery represent major benefits to patients.

Topical anesthetics, intracervical block, and paracervical block

Topical anesthesia, intracervical block (ICB), or paracervical block (PCB) may be offered based on patient preference or the risk of pain, depending on availability at the clinical site (Figure 2). When a PCB is not being placed, the use of certain topical anesthetics or ICB prior to tenaculum placement is encouraged and may be more effective than slow closure of the tenaculum or forced coughing by the patient.^{36,89} Topical anesthetics require a wait time of 5 to 10 minutes (depending on type), whereas ICB/PCBs involve potential injection discomfort. A wait time after ICB/PCB injection is not required as it does not appear to reduce pain (based on abortion studies).^{90–92} Patients should be warned of self-limiting side effects of intravascular lidocaine, such as numbness of the tongue or tinnitus and monitored for symptoms of intravascular lidocaine injection.

Topical or intrauterine anesthetic

Effective options (see Figure 2 for details) include 4 to 5 g EMLA cream applied to the

cervix and cervical canal^{93–96}; 10% lidocaine spray to the cervix and toward the cervical canal^{97,98}; a novel topical formulation of short-acting 4% viscous lidocaine solution⁹⁹; and 10 mL intrauterine mepivacaine 20 mg/mL instilled through a hydrosopy catheter¹⁰⁰ (10% lidocaine and 4% viscous lidocaine are not available commercially in the US). A systematic review and network meta-analysis concluded that EMLA cream (lidocaine 2.5% and prilocaine 2.5%) was the most effective in decreasing pain at tenaculum and IUD placement,⁹⁴ but it has not been studied in nulliparous patients and requires a 7-minute wait.^{96,98} Different formulations of topical intrauterine anesthetics may be useful for reducing IUD placement pain despite evidence of heterogeneous effects.^{20,75,93} Topical options may be especially promising for those wanting to avoid an injection.

Pretreatment with 2% topical lidocaine gel both intravaginally or cervically/intracervically is not effective

in reducing pain with IUD placement.^{101–105} However, self-administered intravaginal 2% lidocaine gel may decrease pain with tenaculum and speculum placement^{101,102} and may be particularly helpful for patients with vestibulodynia, pelvic pain, or a history of trauma.

Intracervical block

In contrast to recent guidance suggesting limited evidence for ICB use,²⁰ we recommend that an ICB be considered as an alternative to PCB and to anesthetize the cervix prior to tenaculum placement. To reduce pain with tenaculum placement, an intracervical injection of 2 mL 1% lidocaine to the cervix immediately prior to tenaculum closure was superior to 2% lidocaine gel applied to the cervix.¹⁰⁶ An ICB with 3.6 mL of 2% lidocaine has also demonstrated reduced pain with tenaculum and IUD placement and improved patient experience in one study of nulliparous patients

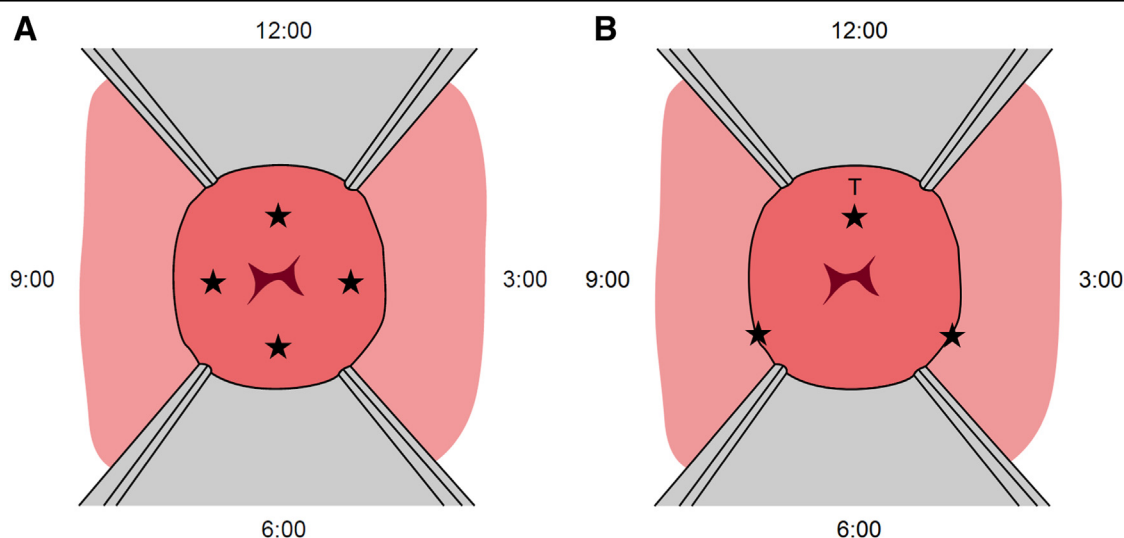
(see Figure 4, A for details of the procedure).¹⁰⁷ Another study, which included nulliparous and parous patients, found ICB with 6 mL of 2% lidocaine to be more effective than naproxen 550 mg alone in reducing LNG-IUD insertion pain.¹⁰⁸ Alternatively, the more commonly available 1% lidocaine and a regular syringe can be used.^{108,109}

Paracervical block

PCB has also been shown to decrease pain with IUD placement in nulliparous patients^{110,111} and may be useful more generally or during challenging IUD placements.²⁵ One study showed that a PCB using 10 mL of 1% lidocaine provided pain relief among nulliparous adolescents (14–25 years old) receiving the 13.5 mg LNG-IUD.¹¹¹ Another study of buffered 20 mL 1% lidocaine among nulliparous patients reported a significant decrease in pain scores during placement, uterine sounding, and 5 minutes after placement; however, block

FIGURE 4

A, Illustration of cervix with ICB injection. B, Illustration of cervix with PCB injection points



A, Illustration of the cervix with ICB injection *Cervical stroma injection sites. Use a 27-gauge needle to inject 4 mL of 2% lidocaine intracervically at the 3, 6, 9, and 12 o'clock positions before tenaculum placement. The original study employed 3.6 mL of 2% lidocaine without vasoconstrictors using a size 6 dental carpule syringe, which may not be available everywhere.¹⁰³ An alternative is to use 1% lidocaine with a standard syringe for an ICB. ICB, intracervical block. **B**, Illustration of cervix with PCB injection points. *Cervicovaginal injection sites. Start with a superficial intracervical injection of 2 mL of buffered 1% lidocaine at the tenaculum site, typically the anterior lip of the cervix, or the posterior lip if there is significant antelexion. After placing the tenaculum, inject the remaining anesthetic at the 4 o'clock and 8:00 o'clock positions at the cervicovaginal junction using a 22- or 25-gauge 1.5" needle.¹⁰⁶ Create a small wheal at the junction and inject slowly while moving the needle to avoid intravascular injection. Aspirating for blood prior to injection can also help.¹⁰²

PCB, paracervical block; T, tenaculum site.

GLOSSARY

CIM complementary and integrative medicine
 ICB intracervical block
 IUD intrauterine device
 LI4 large-intestine 4
 LNG-IUD levonorgestrel IUD
 NSAID nonsteroidal antiinflammatory drug
 PCB paracervical block
 RCT randomized controlled trial
 SP6 spleen-6
 TIC trauma-informed care

administration was more painful than the sham procedure.¹¹⁰ An earlier PCB study of 10 mL buffered 1% lidocaine did not demonstrate significant pain reduction in multiparous and nulliparous patients receiving the 52 mg LNG-IUD or the copper 380 mm² IUD¹¹²; however, the 10 mL dose may have been insufficient given subsequent findings for other procedures.^{113,114} We recommend offering a PCB to nulliparous patients, others at increased risk for pain, and those patients concerned about pain (see Figure 4, B for details of the procedure). PCB is also helpful for more challenging IUD placements, such as those requiring cervical dilation. Buffering the lidocaine by combining 2 mL of sodium bicarbonate with 18 mL of 1% lidocaine for a 20 mL block can decrease the burning sensation during injection. Lack of access to PCB should not prevent IUD placement if indicated; referral to a provider that provides PCB should be made if a patient desires it as part of IUD placement.

Failed first attempt at IUD placement

IUD placement may fail in 1.8% to 20% of the cases.^{83,115,116} Depending on the suspected cause of failure, consider: misoprostol 400 mcg buccally or vaginally 3 to 4 hours before placement or 200 mcg 10 and 4 hours before placement^{83,117}; ultrasound-guided placement⁷³; and/or a PCB/ICB (Figures 4, A and B).

Post-procedure care

After successful placement, the patient is advised to lie supine on the examination table with their legs out of the footrests for 5 minutes. The head of the table is gradually raised to reduce the risk of vasovagal episodes. A drink, snack, and/or heating pad for the lower abdomen can be offered, and acupressure on LI4 or SP6 may be applied bilaterally for a few minutes (Figure 3). The clinician should reinforce the aftercare instructions using the teach-back approach,¹¹⁸ and time should be given to debrief with the patient about their experience and answer any additional questions. The patient is advised to take an NSAID with food on a scheduled basis for the first 24 to 72 hours postprocedure.

Conclusion

This article provides best-practice guidance for managing IUD placement-associated discomfort. Recommendations include a team-based, trauma-informed, personalized, and empowering approach; the use of therapeutic language; optimal environment; non-pharmacological and pharmacological interventions to reduce discomfort; and supporting staff competencies. We encourage all IUD providers to develop relevant skills to support patients who may want or require an IUD. By improving pain management during IUD placement, we can promote a more positive patient experience and reduce the barriers to IUD uptake. ■

ACKNOWLEDGMENTS

The authors would like to thank the original researchers and their patients upon whose evidence these guidelines are built, and Dr Craig Taylor from Highfield Communication Consultancy, Oxford, UK for writing and editorial support.

REFERENCES

1. Abma JC, Martinez GM. Sexual activity and contraceptive use among teenagers in the United States, 2011-2015. *Natl Health Stat Report* 2017;104:1-23.
2. ACOG Committee Opinion No. 735: adolescents and long-acting reversible contraception: implants and intrauterine devices. *Obstet Gynecol* 2018;131:e130-9.

3. Shih G, Vittinghoff E, Steinauer J, Dehlendorf C. Racial and ethnic disparities in contraceptive method choice in California. *Perspect Sex Reprod Health* 2011;43:173-80. <https://doi.org/10.1363/4317311>.
4. Akintomide H, Brima N, Sewell RD, Stephenson JM. Patients' experiences and providers' observations on pain during intrauterine device insertion. *Eur J Contracept Reprod Health Care* 2015;20:319-26.
5. Marions L, Lökvist L, Taube A, Johansson M, Dalvik H, Øverlie I. Use of the levonorgestrel releasing-intrauterine system in nulliparous women—a non-interventional study in Sweden. *Eur J Contracept Reprod Health Care* 2011;16:126-34.
6. Gemzell-Danielsson K, Mansour D, Fiala C, Kaunitz AM, Bahamondes L. Management of pain associated with the insertion of intrauterine contraceptives. *Hum Reprod Update* 2013;19:419-27.
7. Lopes-Garcia EA, Carmona EV, Monteiro I, Bahamondes L. Assessment of pain and ease of intrauterine device placement according to type of device, parity, and mode of delivery. *Eur J Contracept Reprod Health Care* 2023;28:163-7.
8. Chaves IA, Baêta T, Dolabella GB, et al. Pain scores at the insertion of the 52 MG levonorgestrel-releasing intrauterine system among nulligravidas and parous women. *Eur J Contracept Reprod Health Care* 2021;26:399-403.
9. Bracken J, Graham CA. Young women's attitudes towards, and experiences of, long-acting reversible contraceptives. *Eur J Contracept Reprod Health Care* 2014;19:276-84.
10. Michie L, Cameron ST, Glasier A, Wellings K, Loudon J. Myths and misconceptions about intrauterine contraception among women seeking termination of pregnancy. *J Fam Plann Reprod Health Care* 2014;40:36-40.
11. Faculty of Sexual and Reproductive Healthcare. FSRH Statement: Pain Associated With Insertion of Intrauterine Contraception. Edinburgh: Faculty of Sexual and Reproductive Health; 2021.
12. Brown BP, Chor J, Hebert LE, Webb ME, Whitaker AK. Shared negative experiences of long-acting reversible contraception and their influence on contraceptive decision-making: a multi-methods study. *Contraception* 2019;99:228-32.
13. Wu J, Trahair E, Happ M, Swartz J. TikTok, #IUD, and user experience with intrauterine devices reported on social media. *Obstet Gynecol* 2023;141:215-7.
14. Taghinejadi N, van der Westhuizen H-M, Ayomoh FI, Ahmed W, Greenhalgh T, Boylan A-M. Pain experiences during intrauterine device procedures: a thematic analysis of tweets. *BMJ Sex Reprod Health* 2024;50:271-7.
15. Akdemir Y, Karadeniz M. The relationship between pain at IUD insertion and negative perceptions, anxiety and previous mode of delivery. *Eur J Contracept Reprod Health Care* 2019;24:240-5.

16. Dina B, Peipert LJ, Zhao Q, Peipert JF. Anticipated pain as a predictor of discomfort with intrauterine device placement. *Am J Obstet Gynecol* 2018;218:236.e1–9.
17. Maguire K, Morrell K, Westhoff C, Davis A. Accuracy of providers' assessment of pain during intrauterine device insertion. *Contraception* 2014;89:22–4.
18. ACOG. Committee Opinion No 672: clinical challenges of long-acting reversible contraceptive methods. *Obstet Gynecol* 2016;128:e69–77.
19. Daidone C, Morris K, Colquitt J, Jackson G. Provider perspectives on analgesic use in intrauterine device insertion procedures: a mixed methods analysis. *Cureus* 2024;16:e56580.
20. Curtis KM, Nguyen AT, Tepper NK, et al. U. S. Selected practice recommendations for contraceptive use, 2024. *MMWR Recomm Rep* 2024;73:1–77.
21. McCarthy C. Intrauterine contraception insertion pain: nursing interventions to improve patient experience. *J Clin Nurs* 2018;27:9–21.
22. Whiteman MK, Tyler CP, Folger SG, Gaffield ME, Curtis KM. When can a woman have an intrauterine device inserted? A systematic review. *Contraception* 2013;87:666–73.
23. ACOG. ACOG Committee Opinion No. 373: sexual misconduct. *Obstet Gynecol* 2007;110:441–4.
24. Grossman S, Cooper Z, Buxton H, et al. Trauma-informed care: recognizing and resisting re-traumatization in health care. *Trauma Surg Acute Care Open* 2021;6:e000815.
25. ACOG. Caring for patients who have experienced trauma: ACOG Committee Opinion, Number 825. *Obstet Gynecol* 2021;137:e94–9.
26. Dehlendorf C, Levy K, Kelley A, Grumbach K, Steinauer J. Women's preferences for contraceptive counseling and decision making. *Contraception* 2013;88:250–6.
27. ACOG. Informed consent and shared decision making in obstetrics and gynecology: ACOG Committee Opinion, Number 819. *Obstet Gynecol* 2021;137:e34–41.
28. Narayan A, Sheeder J, Guiahi M. Association of anticipated insertional pain with intrauterine device initiation. *J Adolesc Health* 2018;63:37–42.
29. Nicola M, Correia H, Ditchburn G, Drummond PD. Defining pain-validation: the importance of validation in reducing the stresses of chronic pain. *Front Pain Res (Lausanne)* 2022;3:884335.
30. Hillard PJA. Practical tips for intrauterine device counseling, insertion, and pain relief in adolescents: an update. *J Pediatr Adolesc Gynecol* 2019;32:S14–22.
31. Lesko J. How to place an IUD with minimal patient discomfort: managing your patient's pain during IUD placement may make it a more comfortable and satisfying experience for both of you. *OBG Manag* 2023;35:19–25.
32. Gemzell-Danielsson K, Schellschmidt I, Apter D. A randomized, phase II study describing the efficacy, bleeding profile, and safety of two low-dose levonorgestrel-releasing intrauterine contraceptive systems and Mirena. *Fertil Steril* 2012;97:616–22.e3.
33. Wiebe ER. A comparison of the insertion pain associated with three different types of intrauterine device. *Int J Gynaecol Obstet* 2015;129:172.
34. Kaislasuo J, Heikinheimo O, Lähteenmäki P, Suhonen S. Predicting painful or difficult intrauterine device insertion in nulligravid women. *Obstet Gynecol* 2014;124:345–53.
35. Bates CK, Carroll N, Potter J. The challenging pelvic examination. *J Gen Intern Med* 2011;26:651–7.
36. Ireland LD, Allen RH. Pain management for gynecologic procedures in the office. *Obstet Gynecol Surv* 2016;71:89–98.
37. Gemzell-Danielsson K, Jensen JT, Monteiro I, et al. Interventions for the prevention of pain associated with the placement of intrauterine contraceptives: an updated review. *Acta Obstet Gynecol Scand* 2019;98:1500–13.
38. Ngo LL, Ward KK, Mody SK. Ketorolac for pain control with intrauterine device placement: a randomized controlled trial. *Obstet Gynecol* 2015;126:29–36.
39. Ngo LL, Braaten KP, Eichen E, Fortin J, Maurer R, Goldberg AB. Naproxen sodium for pain control with intrauterine device insertion: a randomized controlled trial. *Obstet Gynecol* 2016;128:1306–13.
40. Ross M, Corpin A, Priebe AM, Lawson A. IUD insertion under conscious sedation: patient characteristics with clinical decisions. *J Pediatr Adolesc Gynecol* 2024;37:433–7.
41. Rhudy JL, Meagher MW. Fear and anxiety: divergent effects on human pain thresholds. *Pain* 2000;84:65–75.
42. Nichols M, Goldstein R, Gevirtz C, Healow D. Pain management. In: *Management of unintended and abnormal pregnancy: comprehensive abortion care* 2009;90. Wiley-Blackwell.
43. Allen RH, Singh R. Society of Family Planning clinical guidelines pain control in surgical abortion part 1 — local anesthesia and minimal sedation. *Contraception* 2018;97:471–7.
44. Daykan Y, Battino S, Arbib N, et al. Verbal analgesia is as good as oral tramadol prior to intrauterine device (IUD) insertion, among nulliparous women: a randomized controlled trial. *Eur J Obstet Gynecol Reprod Biol* 2021;258:443–6.
45. Carugno J, Timmons D, Lederer M, Grady MM. Impact of using words with unpleasant emotional connotations on perceived patient discomfort during vaginal speculum examinations: a randomized controlled trial. *Eur J Obstet Gynecol Reprod Biol* 2020;247:203–6.
46. Nilsson U. The anxiety- and pain-reducing effects of music interventions: a systematic review. *AORN J* 2008;87:780–807.
47. Lu X, Thompson WF, Zhang L, Hu L. Music reduces pain unpleasantness: evidence from an EEG study. *J Pain Res* 2019;12:3331–42.
48. Mak N, Reinders IMA, Stockers SA, Westen EHMM, Maas JWM, Bongers MY. The effect of music in gynaecological office procedures on pain, anxiety and satisfaction: a randomized controlled trial. *Gynecol Surg* 2017;14:14.
49. Akin MD, Weingand KW, Hengehold DA, Goodale MB, Hinkle RT, Smith RP. Continuous low-level topical heat in the treatment of dysmenorrhea. *Obstet Gynecol* 2001;97:343–9.
50. Angioli R, De Cicco Nardone C, Plotti F, et al. Use of music to reduce anxiety during office hysteroscopy: prospective randomized trial. *J Minim Invasive Gynecol* 2014;21:454–9.
51. Galaal K, Bryant A, Deane KHO, Al-Khaduri M, Lopes AD. Interventions for reducing anxiety in women undergoing colposcopy. *Cochrane Database Syst Rev* 2011;2011:CD006013.
52. Chan YM, Lee PW, Ng TY, Ngan HY, Wong LC. The use of music to reduce anxiety for patients undergoing colposcopy: a randomized trial. *Gynecol Oncol* 2003;91:213–7.
53. Öz T, Demirci N. The effect of virtual reality glasses applied during intrauterine device insertion on pain, anxiety and satisfaction: randomized controlled study. *Scott Med J* 2024;69:37–44.
54. Benazzouz I, Bouhnik C, Chapron A, Esvan M, Lavoué V, Brun T. Effects of virtual reality on pain during intrauterine device insertions: a randomized controlled trial. *J Gynecol Obstet Hum Reprod* 2024;53:102706.
55. Riska H, Widaryanti R, Yuliani I, Ratnaningsih E. Understanding the effect of virtual reality on anxiety and pain due to intrauterine device insertion. *Iran J Nurs Midwifery Res* 2024;29:140–2.
56. Shahnazi M, Nikjoo R, Yavarikia P, Mohammad-Alizadeh-Charandabi S. Inhaled lavender effect on anxiety and pain caused from intrauterine device insertion. *J Caring Sci* 2012;1:255–61.
57. Alei M, Khazaie F, Rahnama P, et al. Effect of lavender on pain during insertion of intrauterine device: a clinical trial. *J Babol Univ Med Sci* 2013;15:93–9.
58. Abbasijahromi A, Hojati H, Nikooei S, et al. Compare the effect of aromatherapy using lavender and Damask rose essential oils on the level of anxiety and severity of pain following C-section: a double-blinded randomized clinical trial. *J Complement Integr Med* 2020;17. <https://doi.org/10.1515/jcim-2019-0141>.
59. Citlik Saritas S, Buyukbayram Z, Kaplan Serin E, Bilgic Y. Effects of lavender oil intervention before endoscopic retrograde cholangiopancreatography on patients' vital signs, pain and anxiety: a randomized controlled study. *Explore (NY)* 2021;17:446–50.
60. Briggs P, Hawrylack H, Mooney R. Inhaled peppermint oil for postop nausea in patients undergoing cardiac surgery. *Nursing* 2016;46:61–7.
61. Masoumi SZ, Asl HR, Poorolajal J, Panah MH, Oliaei SR. Evaluation of mint efficacy regarding dysmenorrhea in comparison with

mefenamic acid: a double blinded randomized crossover study. *Iran J Nurs Midwifery Res* 2016;21:363–7.

62. Erdoğan P, Yardımcı H. Analgesic effects of LI4 acupuncture during intrauterine device insertion: a randomized controlled clinical trial. *Arch Gynecol Obstet* 2023;308:1361–8.

63. Abaraogu UO, Igwe SE, Tabansi-Ochiogu CS. Effectiveness of SP6 (Sanyinjiao) acupressure for relief of primary dysmenorrhea symptoms: a systematic review with meta- and sensitivity analyses. *Compl Ther Clin Pract* 2016;25:92–105.

64. Passmore R, Gold MA. Nonpharmacologic approaches to pain management with IUD insertion. In: Coles M, Mays A, eds. *Optimizing IUD Delivery for Adolescents and Young Adults*. Cham: Springer; 2019.

65. Dincer Y, Oskay U. The effect of acupressure applied to sanyinjiao (SP6) on primary dysmenorrhea. *Altern Ther Health Med* 2023;29:16–22.

66. Bondi A, Morgan T, Fowler SB. Effects of Reiki on pain and anxiety in women hospitalized for obstetrical- and gynecological-related conditions. *J Holist Nurs* 2021;39:58–65.

67. Chhikara A, Jain M, Vats S, et al. Role of yoga in minimizing stress and anxiety in women experiencing dysmenorrhea. *J Lifestyle Med* 2023;13:90–6.

68. Jayawardena R, Ranasinghe P, Ranawaka H, Gamage N, Dissanayake D, Misra A. Exploring the therapeutic benefits of pranayama (yogic breathing): a systematic review. *Int J Yoga* 2020;13:99–110.

69. Coffin ST, Raj SR. Non-invasive management of vasovagal syncope. *Auton Neurosci* 2014;184:27–32.

70. Hill DA, Lamvu G. Effect of lubricating gel on patient comfort during vaginal speculum examination: a randomized controlled trial. *Obstet Gynecol* 2012;119:227–31.

71. Speedie JMM, Mansour D, Landy R, Clement KM. A randomised trial comparing pain and ease of use of two different stabilising forceps for insertion of intrauterine contraception. *J Fam Plann Reprod Health Care* 2016;42:241–6.

72. Doty N, MacIsaac L. Effect of an atraumatic vulsellum versus a single-tooth tenaculum on pain perception during intrauterine device insertion: a randomized controlled trial. *Contraception* 2015;92:567–71.

73. Baradwan S, Alshahrani MS, Alnoury A, et al. Does ultrasound guidance provide pain relief during intrauterine contraceptive device insertion? A systematic review and meta-analysis of randomized controlled trials. *J Ultrasound Med* 2023;42:1401–11.

74. Allen RH, Bartz D, Grimes DA, Hubacher D, O'Brien P. Interventions for pain with intrauterine device insertion. *Cochrane Database Syst Rev* 2009;2015:CD007373.

75. Lopez LM, Bernholc A, Zeng Y, et al. Interventions for pain with intrauterine device insertion. *Cochrane Database Syst Rev* 2015;2015:CD007373.

76. Grimes DA, Hubacher D, Lopez LM, Schulz KF. Non-steroidal anti-inflammatory drugs for heavy bleeding or pain associated with intrauterine-device use. *Cochrane Database Syst Rev* 2006;2006:CD006034.

77. Karabayirli S, Ayrim AA, Muslu B. Comparison of the analgesic effects of oral tramadol and naproxen sodium on pain relief during IUD insertion. *J Minim Invasive Gynecol* 2012;19:581–4.

78. Crawford M, Davy S, Book N, Elliott JO, Arora A. Oral ketorolac for pain relief during intrauterine device insertion: a double-blinded randomized controlled trial. *J Obstet Gynaecol Can* 2017;39:1143–9.

79. Bednarek PH, Creinin MD, Reeves MF, Cwiak C, Espey E, Jensen JT. Prophylactic ibuprofen does not improve pain with IUD insertion: a randomized trial. *Contraception* 2015;91:193–7.

80. Chor J, Bregand-White J, Golobof A, Harwood B, Cowett A. Ibuprofen prophylaxis for levonorgestrel-releasing intrauterine system insertion: a randomized controlled trial. *Contraception* 2012;85:558–62.

81. Hubacher D, Reyes V, Lillo S, Zepeda A, Chen PL, Croxatto H. Pain from copper intrauterine device insertion: randomized trial of prophylactic ibuprofen. *Am J Obstet Gynecol* 2006;195:1272–7.

82. Swenson C, Turok DK, Ward K, Jacobson JC, Dermish A. Self-administered misoprostol or placebo before intrauterine device insertion in nulliparous women: a randomized controlled trial. *Obstet Gynecol* 2012;120:341–7.

83. Bahamondes MV, Espejo-Arce X, Bahamondes L. Effect of vaginal administration of misoprostol before intrauterine contraceptive insertion following previous insertion failure: a double blind RCT. *Hum Reprod* 2015;30:1861–6.

84. Scavuzzi A, Souza AS, Costa AA, Amorim MM. Misoprostol prior to inserting an intrauterine device in nulligravidas: a randomized clinical trial. *Hum Reprod* 2013;28:2118–25.

85. Bayer LL, Edelman AB, Fu R, et al. An evaluation of oral midazolam for anxiety and pain in first-trimester surgical abortion: a randomized controlled trial. *Obstet Gynecol* 2015;126:37–46.

86. Dhaliwal J, Rosani A, Saadabadi A. *Diazepam StatPearls*. Treasure Island, FL: StatPearls Publishing; 2024.

87. Singh RH, Thaxton L, Carr S, Leeman L, Schneider E, Espey E. A randomized controlled trial of nitrous oxide for intrauterine device insertion in nulliparous women. *Int J Gynaecol Obstet* 2016;135:145–8.

88. Fowler KG, Byraiah G, Burt C, Lee DB, Miller RJ. Nitrous oxide use for intrauterine system placement in adolescents. *J Pediatr Adolesc Gynecol* 2022;35:159–64.

89. Lambert T, Truong T, Gray B. Pain perception with cervical tenaculum placement during intrauterine device insertion: a randomised controlled trial. *BMJ Sex Reprod Health* 2020;46:126–31.

90. Renner RM, Edelman AB, Nichols MD, Jensen JT, Lim JY, Bednarek PH. Refining paracervical block techniques for pain control in first trimester surgical abortion: a randomized controlled noninferiority trial. *Contraception* 2016;94:461–6.

91. Phair N, Jensen JT, Nichols MD. Paracervical block and elective abortion: the effect on pain of waiting between injection and procedure. *Am J Obstet Gynecol* 2002;186:1304–7.

92. Wiebe ER, Rawling M. Pain control in abortion. *Int J Gynaecol Obstet* 1995;50:41–6.

93. Whitworth K, Neher J, Safranek S. Effective analgesic options for intrauterine device placement pain. *Can Fam Physician* 2020;66:580–1.

94. Samy A, Abbas AM, Mahmoud M, et al. Evaluating different pain lowering medications during intrauterine device insertion: a systematic review and network meta-analysis. *Fertil Steril* 2019;111:533–61.e4.

95. Tavakolian S, Ghorbani M, Doulabi MA, Baghban AA, Mortazavi A. Lidocaine-prilocaine cream as analgesia for IUD insertion: a prospective, randomized, controlled, triple blinded study. *Global J Health Sci* 2015;7:p399.

96. Abbas AM, Abdellah MS, Khalaf M, et al. Effect of cervical lidocaine-prilocaine cream on pain perception during copper T380A intrauterine device insertion among parous women: a randomized double-blind controlled trial. *Contraception* 2017;95:251–6.

97. Panichyawat N, Mongkornthong T, Wongwananuruk T, Sirmai K. 10% lidocaine spray for pain control during intrauterine device insertion: a randomised, double-blind, placebo-controlled trial. *BMJ Sex Reprod Health* 2021;47:159–65.

98. Aksoy H, Aksoy Ü, Ozyurt S, Açmaz G, Babayigit M. Lidocaine 10% spray to the cervix reduces pain during intrauterine device insertion: a double-blind randomised controlled trial. *J Fam Plann Reprod Health Care* 2016;42:83–7.

99. Tornblom-Paulander S, Tingåker BK, Werner A, et al. Novel topical formulation of lidocaine provides significant pain relief for intrauterine device insertion: pharmacokinetic evaluation and randomized placebo-controlled trial. *Fertil Steril* 2015;103:422–7.

100. Envall N, Elgemark K, Kopp Kallner H. Mepivacaine instillation for pain reduction during intrauterine device placement in nulliparous women: a double-blinded randomized trial. *Am J Obstet Gynecol* 2024;231:524.e1–7.

101. Conti JA, Lerma K, Schneyer RJ, Hastings CV, Blumenthal PD, Shaw KA. Self-administered vaginal lidocaine gel for pain management with intrauterine device insertion: a blinded, randomized controlled trial. *Am J Obstet Gynecol* 2019;220:177.e1–7.

102. Rapkin RB, Achilles SL, Schwarz EB, et al. Self-administered lidocaine gel for intrauterine device insertion in nulliparous women. *Obstet Gynecol* 2016;128:621–8.

103. Allen RH, Raker C, Goyal V. Higher dose cervical 2% lidocaine gel for IUD insertion: a randomized controlled trial. *Contraception* 2013;88:730–6.

- 104.** Maguire K, Davis A, Rosario Tejeda L, Westhoff C. Intracervical lidocaine gel for intrauterine device insertion: a randomized controlled trial. *Contraception* 2012;86:214–9.
- 105.** McNicholas CP, Madden T, Zhao Q, Secura G, Allsworth JE, Peipert JF. Cervical lidocaine for IUD insertional pain: a randomized controlled trial. *Am J Obstet Gynecol* 2012;207:384.e1–6.
- 106.** Goldthwaite LM, Baldwin MK, Page J, et al. Comparison of interventions for pain control with tenaculum placement: a randomized clinical trial. *Contraception* 2014;89:229–33.
- 107.** De Nadai MN, Poli-Neto OB, Franceschini SA, et al. Intracervical block for levonorgestrel-releasing intrauterine system placement among nulligravid women: a randomized double-blind controlled trial. *Am J Obstet Gynecol* 2020;222:245.e1–10.
- 108.** de Oliveira ECF, Baêta T, Brant APC, Silva-Filho A, Rocha ALL. Use of naproxen versus intracervical block for pain control during the 52-mg levonorgestrel-releasing intrauterine system insertion in young women: a multivariate analysis of a randomized controlled trial. *BMC Womens Health* 2021;21:377.
- 109.** Beecham GB, Nessel TA, Goyal A. Lidocaine. In: StatPearls. Treasure Island, FL: StatPearls Publishing LLC; 2022.
- 110.** Mody SK, Farala JP, Jimenez B, Nishikawa M, Ngo LL. Paracervical block for intrauterine device placement among nulliparous women: a randomized controlled trial. *Obstet Gynecol* 2018;132:575–82.
- 111.** Akers AY, Steinway C, Sonalkar S, et al. Reducing pain during intrauterine device insertion: a randomized controlled trial in adolescents and young women. *Obstet Gynecol* 2017;130:795–802.
- 112.** Mody SK, Kiley J, Rademaker A, Gawron L, Stika C, Hammond C. Pain control for intrauterine device insertion: a randomized trial of 1% lidocaine paracervical block. *Contraception* 2012;86:704–9.
- 113.** Renner RM, Nichols MD, Jensen JT, Li H, Edelman AB. Paracervical block for pain control in first-trimester surgical abortion: a randomized controlled trial. *Obstet Gynecol* 2012;119:1030–7.
- 114.** Soon R, Tschann M, Salcedo J, Stevens K, Ahn HJ, Kaneshiro B. Paracervical block for laminaria insertion before second-trimester abortion: a randomized controlled trial. *Obstet Gynecol* 2017;130:387–92.
- 115.** Wise MK, Okuyemi O, Flint M, et al. Intrauterine device placement success for adolescents and young adults at community-based reproductive health clinics. *J Pediatr Adolesc Gynecol* 2024;37:160–4.
- 116.** Dermish AI, Turok DK, Jacobson JC, Flores ME, McFadden M, Burke K. Failed IUD insertions in community practice: an unrecognized problem? *Contraception* 2013;87:182–6.
- 117.** Zapata LB, Jatlaoui TC, Marchbanks PA, Curtis KM. Medications to ease intrauterine device insertion: a systematic review. *Contraception* 2016;94:739–59.
- 118.** Yen PH, Leasure AR. Use and effectiveness of the teach-back method in patient education and health outcomes. *Fed Pract* 2019;36:284–9.
- 119.** Miles SM, Shvartsman K, Dunlow S. Intrauterine lidocaine and naproxen for analgesia during intrauterine device insertion: randomized controlled trial. *Contracept Reprod Med* 2019;4:13.
- 120.** Isley MM, Jensen JT, Nichols MD, Lehman A, Bednarek P, Edelman A. Intrauterine lidocaine infusion for pain management during outpatient transcervical tubal sterilization: a randomized controlled trial. *Contraception* 2012;85:275–81.
- 121.** Nelson AL, Fong JK. Intrauterine infusion of lidocaine does not reduce pain scores during IUD insertion. *Contraception* 2013;88:37–40.
- 122.** Salcedo J, Cleland K, Bartz D, Thompson I. Society of family planning clinical recommendation: emergency contraception. *Contraception* 2023;121:109958.
- 123.** Ramanadhan S, Jensen J. The levonorgestrel-releasing intrauterine device as emergency contraception: Re-examining the data. *Obstet Gynecol* 2024;143:189–94.
- 124.** Forouhari S, Piraloo Z, Ghaemi SZ, et al. Analgesia and anesthesia during IUD insertion: local anesthetics. *Int Electron J Med* 2014;3:25–38.
- 125.** Santos ARG, Bahamondes MV, Hidalgo MM, Atti A, Bahamondes L, Monteiro I. Pain at insertion of the levonorgestrel-releasing intrauterine system in nulligravida and parous women with and without cesarean section. *Contraception* 2013;88:164–8.
- 126.** Allen RH, Carey MS, Raker C, Goyal V, Matteson K. A prospective cohort study of pain with intrauterine device insertion among women with and without vaginal deliveries. *J Obstet Gynaecol* 2014;34:263–7.
- 127.** Sinning KM, Jude DC, Yoost JL. Post-insertional pain after intrauterine device placement among nulliparous adolescents. *J Pediatr Adolesc Gynecol* 2018;31:400–4.